

Global Aviation Industry - Business Questions

business questions

- 1.Which airline earned the highest revenue?
 - 2.Which region recovered fastest after COVID?
 - 3.Did Boeing lose market share to Airbus?
 4. Which routes generate the most revenue?
 - 5.Which business model has the highest operating margin?
 6. Has aviation become safer over time?
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Business Scenario Questions

1. Revenue Performance

Business Scenario:

The CEO wants to know which airlines generated the highest revenue over the last 17 years.

Question:

Which airlines generated the highest total revenue between 2010 and 2026?

2. COVID Impact

Business Scenario:

The management team wants to measure the financial damage caused by COVID-19.

Question:

Which airlines experienced the largest revenue decline from 2019 to 2020?

3. Recovery Analysis

Business Scenario:

Executives want to identify which airlines recovered the fastest after the pandemic.

Question:

Which airlines returned to or exceeded their 2019 revenue levels first?

4. Operating Profitability

Business Scenario:

The CFO wants to evaluate operational efficiency.

Question:

Which airlines consistently maintained the highest operating margins?

5. Business Model Comparison

Business Scenario:

Management wants to compare airline business models.

Question:

How do legacy, low-cost, and hybrid airlines compare in revenue growth and operating margins?

6. Passenger Growth

Business Scenario:

The commercial team wants to understand demand growth.

Question:

Which airlines carried the highest number of passengers over the study period?

7. Load Factor Analysis

Business Scenario:

Operations wants to improve seat utilization.

Question:

Which airlines consistently achieved the highest passenger load factors?

8. Regional Recovery

Business Scenario:

Executives want to compare recovery across world regions.

Question:

Which region recovered fastest after COVID, and which recovered the slowest?

9. Manufacturer Competition

Business Scenario:

Investors want to compare aircraft manufacturers.

Question:

How has Boeing's market position changed compared to Airbus and COMAC between 2010 and 2026?

10. Boeing Crisis Analysis

Business Scenario:

The board wants to understand the impact of the 737 MAX crisis.

Question:

How did the Boeing 737 MAX crisis affect aircraft orders, deliveries, and backlog?

11. Fleet Demand**Business Scenario:**

Aircraft manufacturers want to forecast production.

Question:

Which aircraft families received the highest number of net orders over the study period?

12. Route Profitability**Business Scenario:**

Network planners want to maximize profitability.

Question:

Which flight routes generated the highest annual revenue?

13. Fare Analysis**Business Scenario:**

Revenue management wants pricing insights.

Question:

Which routes have the highest average fare, and how does fare relate to passenger volume?

14. Traffic Trends**Business Scenario:**

Industry analysts want to understand travel demand.

Question:

How did passenger traffic (RPK) change across different regions from 2010 to 2026?

15. Aviation Safety**Business Scenario:**

The safety committee wants to review historical incident trends.

Question:

Has the number of major aviation incidents increased or decreased over time?

16. Manufacturer Safety Analysis

Business Scenario:

Executives want to compare aircraft safety records.

Question:

Were Boeing aircraft involved in more major incidents than Airbus aircraft during the study period?

17. Geopolitical Impact

Business Scenario:

Risk management wants to assess geopolitical disruptions.

Question:

How have geopolitical events affected airlines and aviation operations?

18. Middle East Aviation Growth

Business Scenario:

Investors are evaluating Middle Eastern airlines.

Question:

How have Emirates, Qatar Airways, Etihad, Saudia, and Riyadh Air grown compared to other global airlines?

19. Future Growth Opportunities

Business Scenario:

Executives want to identify future investment opportunities.

Question:

Which airlines, manufacturers, or regions show the strongest long-term growth trends?

20. Executive Dashboard

Business Scenario:

The CEO needs a single dashboard to monitor business performance.

Question:

Which KPIs should be included in an executive aviation industry dashboard to monitor financial performance, passenger demand, operational efficiency, fleet activity, and safety?

Bonus: Advanced Business Questions (Great for Portfolio)

1. Which airlines achieved the highest revenue per passenger?
2. Which regions generated the highest passenger demand after COVID?
3. Is there a relationship between fleet size and revenue growth?
4. Do higher load factors lead to higher operating margins?
5. Which aircraft manufacturer maintained the strongest order backlog during industry crises?
6. How did major aviation incidents affect airline performance in subsequent years?
7. Which routes recovered passenger demand the fastest after COVID?
8. How has COMAC's emergence affected the Boeing–Airbus market share?
9. Which airlines demonstrated the greatest financial resilience during industry disruptions?
10. What are the key drivers of airline profitability based on revenue, passengers, fleet size, and load factor?

These questions cover descriptive, diagnostic, comparative, and strategic analysis, making them ideal for demonstrating end-to-end data analysis skills in SQL, PostgreSQL, Power BI, Excel, and DAX.

Revenue & Financial Analysis (1–10)

1. Which airline generated the highest total revenue from 2010–2026?
 2. What was the year-over-year revenue growth for each airline?
 3. Which airlines experienced the largest revenue decline during COVID-19?
 4. Which airlines recovered to pre-COVID (2019) revenue levels the fastest?
 5. What is the average annual revenue by airline?
 6. Which region generated the highest airline revenue?
 7. Which airline had the highest operating income over the study period?
 8. Which airlines reported negative operating margins, and in which years?
 9. What percentage of total industry revenue did the top 5 airlines contribute?
 10. Which years had the highest and lowest industry revenue?
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Passenger & Demand Analysis (11–20)

11. Which airline carried the most passengers?
12. How did passenger traffic change year over year?
13. Which region experienced the highest passenger demand?
14. Which region had the fastest post-COVID recovery?

15. Which region had the highest average load factor?
 16. Which airlines consistently maintained a load factor above 85%?
 17. Which month has the highest passenger demand across all regions?
 18. Which region had the steepest decline in RPK during COVID?
 19. Compare RPK vs ASK for each region.
 20. Which airlines improved their load factor the most after COVID?
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Fleet & Manufacturer Analysis (21–30)

21. Which manufacturer received the highest gross orders?
 22. Which manufacturer had the highest net orders?
 23. How did Boeing's orders change after the 737 MAX crisis?
 24. Compare Boeing and Airbus deliveries by year.
 25. Which aircraft family had the highest demand?
 26. Which manufacturer has the largest backlog?
 27. How did COMAC grow between 2023 and 2026?
 28. What percentage of all aircraft orders belongs to Boeing vs Airbus?
 29. Which years saw the highest aircraft deliveries?
 30. Which manufacturer experienced the largest order cancellations?
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Regional Analysis (31–40)

31. Which region generated the highest passenger traffic?
 32. Which region recovered first after COVID?
 33. Which region had the highest average revenue growth?
 34. Compare North America and Europe over time.
 35. Which region had the lowest average load factor?
 36. Which regions show the strongest long-term growth?
 37. Which region experienced the largest COVID decline?
 38. How has Asia Pacific recovered since 2020?
 39. Which regions exceeded 2019 traffic levels by 2026?
 40. Rank all regions by passenger growth.
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Route Performance Analysis (41–50)

41. Which routes generated the highest annual revenue?
 42. Which routes had the highest passenger volume?
 43. Which routes had the highest average fare?
 44. Is there a relationship between route distance and fare?
 45. Which routes recovered fastest after COVID?
 46. Which routes generate the highest revenue per passenger?
 47. Which regions contain the most profitable routes?
 48. Which airlines dominate the top 10 routes?
 49. Which routes have the highest weekly frequency?
 50. Does route distance influence passenger demand?
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Aviation Safety Analysis (51–60)

51. How many major aviation incidents occurred each year?
 52. Which airline experienced the most incidents?
 53. Which manufacturer (Boeing vs Airbus) had more incidents?
 54. Which years recorded the highest fatalities?
 55. How many incidents were geopolitical?
 56. What percentage of incidents were fatal?
 57. Compare fatal and non-fatal incidents by year.
 58. Which aircraft models appear most frequently in incidents?
 59. Has aviation become safer over time?
 60. Which regions experienced the most major incidents?
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Advanced Business Questions (61–75)

61. Which airlines generated the highest revenue per passenger?
62. Which airlines achieved the highest revenue per aircraft?
63. Does fleet size correlate with revenue growth?
64. Is there a relationship between load factor and operating margin?
65. Which airlines were the most resilient during COVID?
66. Which airlines showed the fastest recovery after COVID?

- 67. Which business model (Legacy vs Low-Cost vs Hybrid) performed best?
 - 68. Which airlines consistently ranked in the top 5 by revenue?
 - 69. Which airlines improved profitability the most over time?
 - 70. Which regions offer the greatest growth opportunities?
 - 71. Which airline had the highest compound annual growth rate (CAGR)?
 - 72. Which years had the greatest industry volatility?
 - 73. How did major aviation incidents affect passenger traffic?
 - 74. Which manufacturers gained market share after Boeing's crisis?
 - 75. Which KPIs best measure airline performance?
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Power BI Dashboard Questions (76–90)

These can become dashboard pages.

Executive Overview

- 76. What is the total industry revenue?
- 77. How many passengers traveled globally?
- 78. What is the average load factor?
- 79. Which airline leads in revenue?
- 80. Which manufacturer leads in aircraft orders?

Financial Dashboard

- 81. Revenue trend by year
- 82. Operating margin trend
- 83. Top 10 airlines by revenue
- 84. Revenue by business model
- 85. Revenue by region

Passenger Dashboard

- 86. Passenger trend
 - 87. Load factor trend
 - 88. Regional traffic comparison
 - 89. Monthly traffic trend
 - 90. Recovery timeline by region
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Advanced SQL Challenge Questions (91–100)

These require Window Functions, CTEs, Subqueries, and advanced SQL.

91. Rank airlines by annual revenue using RANK().
 92. Find the top 3 airlines by revenue in each year.
 93. Calculate Month-over-Month passenger growth using LAG().
 94. Calculate Year-over-Year revenue growth.
 95. Find airlines with revenue growth for at least 5 consecutive years.
 96. Identify years when Boeing orders fell below the previous year.
 97. Calculate cumulative revenue by airline using a window function.
 98. Find airlines whose revenue is above the industry average.
 99. Calculate each airline's market share by year.
 100. Build an executive KPI summary combining all five tables.
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13. Business Value

The DAX layer transforms the raw aviation data into meaningful business metrics.

Instead of simply displaying raw columns such as revenue, passengers, or orders, the measures allow stakeholders to answer questions such as:

- How much revenue did airlines generate?
 - How quickly did revenue change?
 - How many passengers were carried?
 - How efficiently was capacity utilized?
 - How many aircraft were ordered?
 - How large is the aircraft backlog?
 - Which airlines are leading by revenue?
 - How severe were aviation incidents?
 - How did performance change during COVID-19?
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11.10 Business Questions Addressed

The dashboards were designed to answer important business questions such as:

1. How has aviation industry revenue changed from 2010–2026?
2. How did COVID-19 affect airline revenue and passenger traffic?
3. Which airlines generated the highest revenue?
4. Which business models performed better?

5. Which regions recovered fastest after COVID-19?
 6. How did Boeing's orders change during the 737 MAX crisis?
 7. How does Boeing compare with Airbus in orders and backlog?
 8. Is COMAC becoming a significant competitor?
 9. Which global routes have the highest passenger volumes?
 10. Which routes generate the highest estimated revenue?
 11. What is the relationship between fares and passenger volume?
 12. How have major incidents changed over time?
 13. Which years recorded the highest fatalities?
 14. How do selected Boeing and Airbus incidents compare?
 15. What role do geopolitical events play in the selected incidents?
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Dashboard 1 — Executive Overview

1. Revenue Performance

How has total airline revenue changed from 2010 to 2026, and which years experienced the largest increases or declines?

2. Passenger Growth

How has the number of passengers carried changed over time, and does passenger growth correspond with revenue growth?

3. Regional Revenue

Which regions generate the highest airline revenue, and how has their contribution changed over time?

4. Airline Leadership

Which 10 airlines generated the highest total revenue during the selected period?

5. Business Model Performance

How does airline performance differ across legacy, low-cost, and hybrid business models?

Dashboard 2 — Airline Performance

6. Airline Revenue Ranking

Which airlines have the highest revenue, and what factors may explain their stronger financial performance?

7. Operating Margin

Which airlines achieved the highest operating margins, and how stable were their margins over time?

8. COVID Impact on Airlines

Which airlines experienced the largest revenue and operating-margin decline during 2020–2021?

9. Fleet & Revenue

Is there a relationship between fleet size and airline revenue?

10. Revenue vs Passengers

Which airlines generate relatively high revenue despite carrying fewer passengers, and what does this suggest about their pricing or business model?

Dashboard 3 — Boeing vs Airbus

11. Manufacturer Competition

How did Boeing, Airbus, COMAC, and Embraer compare in aircraft orders from 2010–2026?

12. Boeing 737 MAX Impact

How significantly did Boeing's aircraft orders change during the 2019–2020 period compared with previous years?

13. Orders vs Deliveries

Which manufacturer converted aircraft orders into deliveries most effectively over the period?

14. Backlog

Which manufacturer has the largest aircraft backlog, and how has the competitive gap changed over time?

15. Aircraft Demand

Which aircraft families generated the highest number of orders?

16. Narrowbody Competition

How has the narrowbody aircraft market evolved between Boeing, Airbus, and COMAC?

Dashboard 4 — Passenger & Route Analysis

17. COVID Regional Impact

Which region experienced the largest decline in passenger traffic during the COVID-19 period?

18. Recovery Speed

Which regions recovered to approximately their 2019 passenger-traffic levels first, and which regions took the longest?

19. Load Factor

How did regional passenger load factors change before, during, and after COVID-19?

20. Top Global Routes

Which routes carried the highest number of passengers during the analyzed period?

21. Route Revenue

Which global routes generated the highest estimated annual revenue?

22. Route Economics

Which routes combine high passenger volume with high average fares and therefore represent potentially attractive routes?

Dashboard 5 — Aviation Safety

23. Incident Trends

How has the number of major aviation incidents in the dataset changed from 2010–2026?

24. Fatalities

Which years experienced the highest number of fatalities, and what major incidents contributed to those totals?

25. Boeing vs Airbus Incidents

How do the numbers of selected major incidents involving Boeing and Airbus aircraft compare?

26. Incident Severity

What proportion of the selected incidents were fatal crashes, fatal collisions, operational incidents, ground incidents, or geopolitical events?

27. Geopolitical Risk

How significant were geopolitical events within the selected aviation incidents?

28. Aircraft Risk

Which aircraft types were associated with the highest number of fatalities in the selected incidents?

10 Most Important Questions for Your Final Presentation

"What business problems did you solve with this project?"

Don't mention all 28 questions. Focus on these **10**:

- 1. How did COVID-19 affect airline revenue and passenger demand globally?**
- 2. Which regions recovered fastest after the COVID-19 disruption?**
- 3. Which airlines demonstrated the strongest financial recovery?**
- 4. Which business model-legacy, low-cost, or hybrid-performed best?**

5. **How did Boeing's 737 MAX crisis affect its aircraft orders compared with Airbus?**
 6. **How has Boeing's position compared with Airbus in aircraft orders, deliveries, and backlog?**
 7. **Is COMAC emerging as a meaningful competitor to Boeing and Airbus?**
 8. **Which global routes have the highest passenger volume and revenue potential?**
 9. **How has the frequency and severity of major aviation incidents changed over time in this selected dataset?**
 10. **What are the major strategic trends shaping the global aviation industry from 2010–2026?**
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